Recap: Lecture 29

In Lecture 29, we

- Introduced a FT table
- Introduced convolution and shift theorems for FT

$$\mathscr{F}\{f * g\} = \widehat{f}(k)\,\widehat{g}(k), \quad \mathscr{F}\{f(x-y)\} = \mathrm{e}^{-iky}\,\widehat{f}(k) \;\; (y \in \mathbb{R}),$$

$$\mathscr{F}\{\mathrm{e}^{iqx} f\} = \widehat{f}(k-q) \;\; (q \in \mathbb{R})$$

- Learnt rules to help apply FTs to ODEs

$$\mathscr{F}\left(f^{(n)}\right) = (ik)^n \widehat{f}, \quad \mathscr{F}(x^n f) = i^n \frac{\mathrm{d}^n \widehat{f}}{\mathrm{d}k^n}$$

Lecture 30

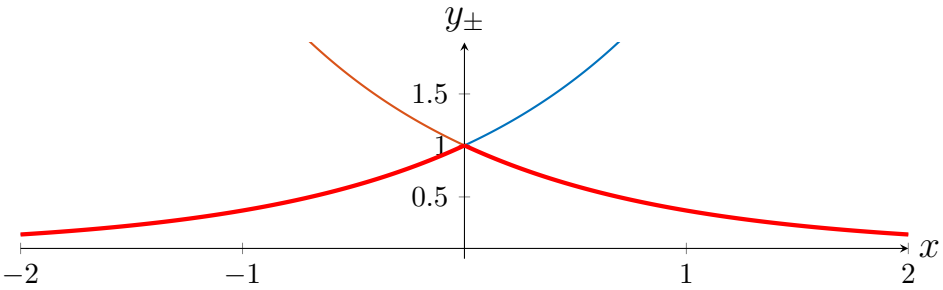
In Lecture 30, we will:

1. Apply the FT to the Airy ODE
2. Apply the FT to a PDE (1D heat equation)

Exercise 6.9 Revisited

$$y''(x) - y(x) = -2\,\delta(x) \quad \longrightarrow \quad \text{FT} \quad \longrightarrow \quad y(x) = \frac{-1}{2}\mathrm{e}^{-|x|}$$

$$y''(x) - y(x) = 0 \quad \Rightarrow \quad y(x) = C_+\, y_+(x) + C_-\, y_-(x) \quad \text{where} \quad y_\pm(x) = \mathrm{e}^{\pm x}$$



Recap (Topic 2): Airy's equation $y''(x) - x\,y(x) = 0$

$$\boxed{\text{Has no solutions in terms of elementary functions}}$$

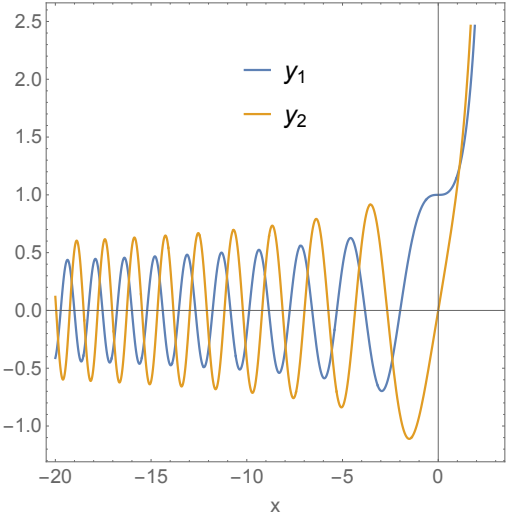
Two independent series solutions:

$$y_1(x) = 1 + \sum_{k=1}^{\infty} \frac{x^{3k}}{2 \cdot 3 \cdot 5 \cdot 6 \cdots (3k-1) \cdot 3k}$$

$$y_2(x) = x + \sum_{k=1}^{\infty} \frac{x^{3k+1}}{3 \cdot 4 \cdot 6 \cdot 7 \cdots 3k \cdot (3k+1)}$$

The general solution of Airy's eqn.

$$y(x) = C_1\, y_1(x) + C_2\, y_2(x)$$



Recap (Topic 2): Airy functions

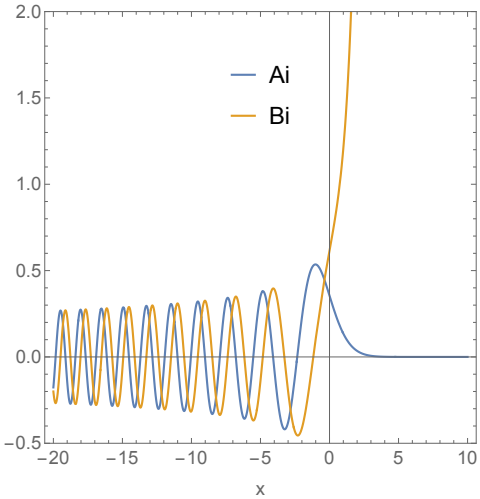
The *Airy functions* are

$$\mathrm{Ai}(x) = \alpha\left[y_1(x) - \beta y_2(x)\right]$$

$$\mathrm{Bi}(x) = \sqrt{3}\alpha\left[y_1(x) + \beta y_2(x)\right]$$

where $\alpha \approx 0.355$ and $\beta \approx 0.729$.

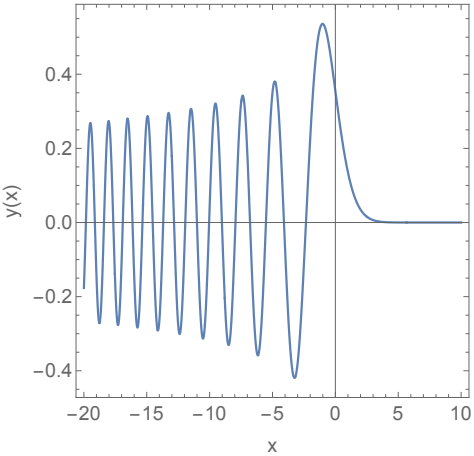
Note Ai(x) $\to 0$ as $x \to \infty$



Exercise 6.10: Use FTs to solve Airy's equation $y''(x) - xy(x) = 0$

We have found Ai in integral form

$$y(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}\,(k^3/3 + k\,x)}\ \mathrm{d}k$$



Transforms applied to PDEs

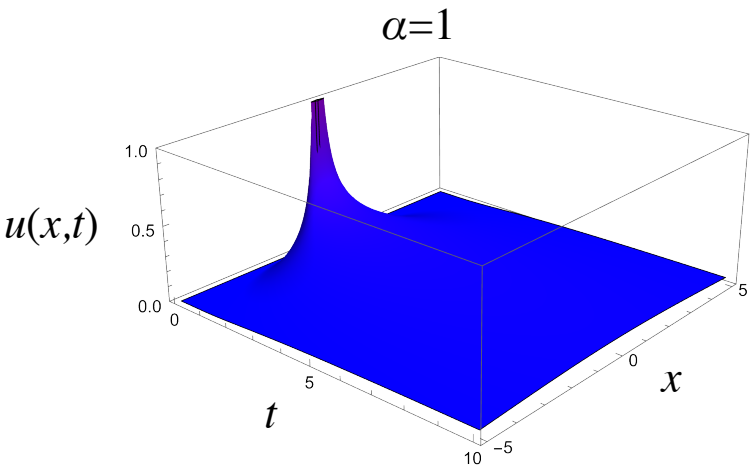
- We've already seen that Laplace transforms and Fourier transforms turn ODEs into algebraic equations
- Transforms can also be used to turn PDEs into ODEs
- Let $u(x,t)$ be defined for $x \in \mathbb{R}$ and $t > 0$.
  - Let $\widehat{u}(k,t) = \mathscr{F}\{u(x,t)\}$
  - $\mathscr{F}\left(\dfrac{\partial^n u}{\partial x^n}\right) = (ik)^n\,\widehat{u}(k,t)$, etc.

Exercise 6.11: 1D Heat Equation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \quad \text{where} \quad \alpha > 0, \quad x \in \mathbb{R}, \quad t > 0,$$

subject to the initial condition $u(x, 0) = f(x)$ and $u(x, t) \to 0$ for $x \to \pm\infty$.

Find the particular solution for $f(x) = \delta(x)$.


α=1
u(x,t)
1.0
0.5
0.0
0
5
10
t
−5
0
5
x